

Packet Tracer – Communicating in a Cyber World

Addressing Table

Device	Private IP Address	Public IP Address	Subnet Mask	Site
FTP/Web Server	10.44.1.254	209.165.201.3	255.255.255.0	Metropolis Bank HQ
Email/DNS Server	10.44.1.253	209.165.201.4	255.255.255.0	Metropolis Bank HQ
NTP/AAA Server	10.44.1.252	209.165.201.5	255.255.255.0	Metropolis Bank HQ
File Backup Server	10.44.2.254	N/A	255.255.255.0	Gotham Healthcare Branch

Objectives

Part 1: Send Email between Users

Part 2: Upload and Download Files using FTP

Part 3: Remotely Access an Enterprise Router using Telnet

Part 4: Remotely Access an Enterprise Router using SSH

Background

In this activity, you will communicate across remote networks using common network services. The IP addressing, network configuration, and service configurations are already complete. You will use the client devices in the differing geographic regions to connect to both servers and other client devices.

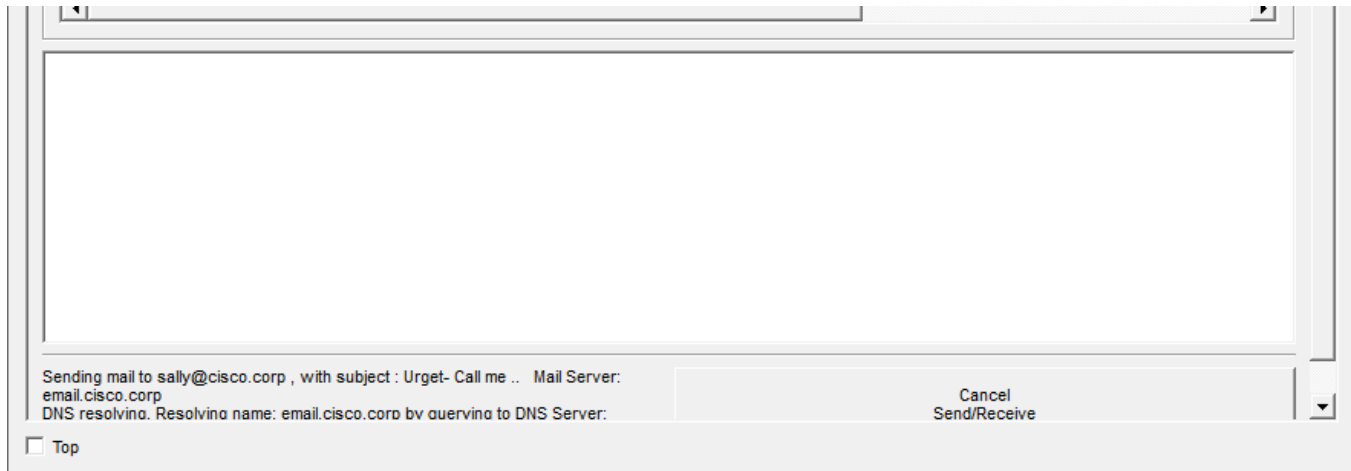
Part 1: Send Email between Users

Step 1: Access the email client on Mike's PC.

- Click the **Gotham Healthcare Branch** site and then click the PC **Mike**.
- Click the **Desktop** tab and then click **Email**.

Step 2: Send an email to Sally.

- Create an email by clicking the **Compose** button.
- In the **To:** field, enter the email sally@cisco.corp
In the **Subject:** field, enter the string of text "**Urgent- Call me**".
In the **Message** section, enter. "**Call me when you are free today to discuss the new sale.**"
- Click the **Send** button to transmit the email.
What protocol was used to send the email to the email server?



Step 3: Have Sally check her email.

- Enter the **Metropolis Bank HQ** site and then click the PC **Sally**.
 - Click the **Desktop** tab and then click **Email**.
 - Click the **Receive** button to retrieve the email sent from Mike.
- What protocol was used to retrieve the email from the email server?



Part 2: Upload Files using FTP

Step 1: Set the packet sniffer to capture traffic on the correct port.

- Enter the geographic (root) view to see all three remote sites.
- Click the **Cyber Criminals Sniffer**.
- Click **Port1** to capture packets on this port.
- Leave the **Cyber Criminal Sniffer** open and visible for the rest of this part.

Step 2: Remotely connect to the FTP server.

- e. Enter the **Healthcare at Home** site and then click the PC **Mary**.
- f. Click the **Desktop** tab and then click **Command Prompt**.
- g. Connect to the **FTP/Web** server at **Metropolis Bank HQ** by entering **ftp 209.165.201.3** in the command prompt.
- h. Enter the username of **mary** and a password of **cisco123**.

Step 3: Upload a file to the FTP server.

- a. At the **ftp>** prompt, enter the command **dir** to view the current files stored on the remote FTP server.
- b. Mary has a file containing sensitive information regarding new healthcare client information.
Upload the **newclients.txt** file to the FTP server by entering the command **put newclients.txt**.
- c. At the **ftp>** prompt, enter the command **dir** and verify the **newclients.txt** file is now on the FTP server.
Why is FTP considered an insecure protocol for moving files?

```
Writing file newclients.txt to 209.165.201.3:
File transfer in progress...

[Transfer complete - 644 bytes]

644 bytes copied in 0.101 secs (6376 bytes/sec)
ftp>dir

Listing /ftp directory from 209.165.201.3:
 0  : BankData.txt                1553
 1  : asa842-k8.bin              5571584
 2  : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 3  : c1841-ipbase-mz.123-14.T7.bin 13832032
 4  : c1841-ipbasek9-mz.124-12.bin 16599160
 5  : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 6  : c2600-i-mz.122-28.bin      5571584
 7  : c2600-ipbasek9-mz.124-8.bin 13169700
 8  : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
 9  : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
10  : c2800nm-ipbase-mz.123-14.T7.bin 5571584
11  : c2800nm-ipbasek9-mz.124-8.bin 15522644
12  : c2950-i6q4l2-mz.121-22.EA4.bin 3058048
13  : c2950-i6q4l2-mz.121-22.EA8.bin 3117390
14  : c2960-lanbase-mz.122-25.FX.bin 4414921
15  : c2960-lanbase-mz.122-25.SEE1.bin 4670455
16  : c2960-lanbasek9-mz.150-2.SE4.bin 4670455
17  : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
18  : newclients.txt            644
19  : pt1000-i-mz.122-28.bin    5571584
20  : pt3000-i6q4l2-mz.121-22.EA4.bin 3117390
ftp>
```

Step 4: Analyze the FTP traffic.

- a. Enter the geographic (root) view to see all three remote sites.
- b. Click the **Cyber Criminals Sniffer**.
- c. Under the GUI tab on the left, click the 1st FTP packet available to select it. Then scroll down to the bottom of the window displayed on the right.

What information is displayed in clear text from the FTP header?

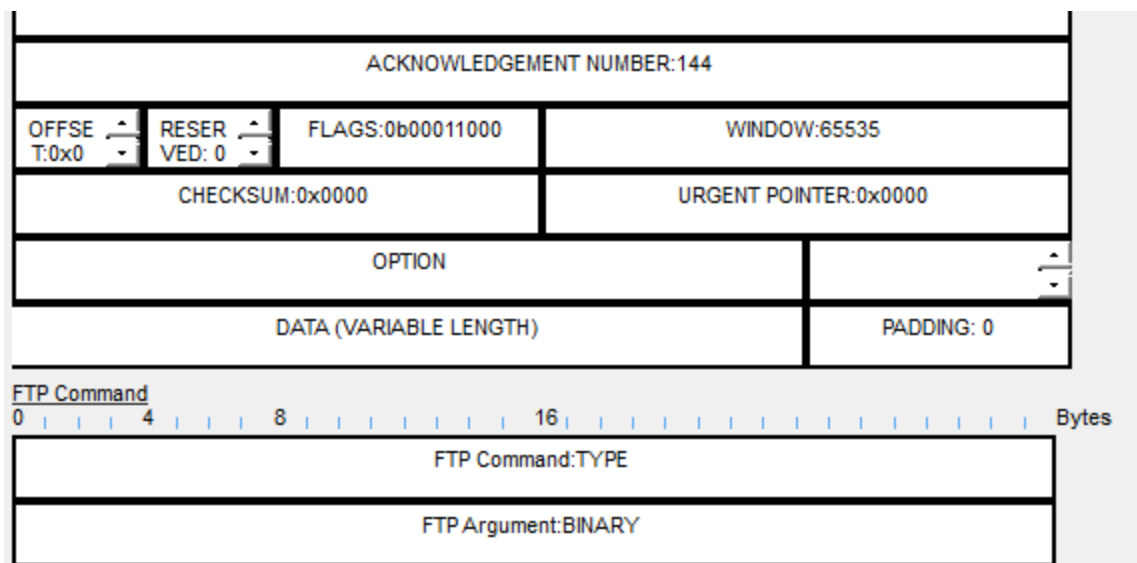
ACKNOWLEDGEMENT NUMBER:53			
OFFSE T:0x0	RESER VED: 0	FLAGS:0b00011000	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

FTP Command			
0	4	8	16
Bytes			
FTP Command:USER			
FTP Argument:mary			

- d. On the left, click the 2nd FTP packet available to select it. Then scroll down to the bottom of the window displayed on the right. Do this again for the 3rd FTP packet.

ACKNOWLEDGEMENT NUMBER:107			
OFFSE T:0x0	RESER VED: 0	FLAGS:0b00011000	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

FTP Command			
0	4	8	16
Bytes			
FTP Command:PASS			
FTP Argument:cisco123			



- e. Besides the username, what other sensitive information is displayed in clear text from the FTP header?

Part 3: Remotely Access an Enterprise Router Using Telnet

Step 1: Remotely connect to an enterprise router.

- Enter the **Healthcare at Home** site and then click on the PC **Dave**.
- Click the **Desktop** tab and then click **Command Prompt**.
- Ping the enterprise router using the command **ping 209.165.201.2** to verify reachability.
- Use the command **telnet 209.165.201.2** to telnet to the IP address of the enterprise router.
- Authenticate to the enterprise router with the username of **admin** and the password of **cisco123**.
- Use the command **show users** to view the active Telnet connection to the enterprise router.

Why is Telnet considered an insecure protocol for remotely managing a device?

```
Minimum = 91ms, Maximum = 117ms, Average = 100ms

C:\>telnet 209.165.201.2
Trying 209.165.201.2 ...Open

User Access Verification

Username: admin
Password:
% Login invalid

Username: admin
Password:
% Password: timeout expired!

[Connection to 209.165.201.2 closed by foreign host]
C:\>telnet 209.165.201.2
Trying 209.165.201.2 ...Open

User Access Verification

Username: admin
Password:
HQ_Router#show users
  Line      User      Host(s)      Idle      Location
*390 vty 0   admin     idle         00:00:00  209.165.201.11

  Interface  User      Mode      Idle      Peer Address
```

Part 4: Remotely Access an Enterprise Router Using SSH

Step 1: Remotely connect to an enterprise router.

- Enter the **Gotham Healthcare Branch** site and then click the PC **Tim**.
- Click the **Desktop** tab and then click **Command Prompt**.
- Ping the enterprise router using the command **ping 209.165.201.2** to verify reachability.
- Use the command **ssh -l admin 209.165.201.2** to SSH to the IP address of the enterprise router.
- Authenticate to the enterprise router with the password of **cisco123**.
- Use the command **show users** to view the active SSH connection to the enterprise router.

Why is SSH considered a secure protocol for remotely managing a device?

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.201.2

Pinging 209.165.201.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 209.165.201.2: bytes=32 time=146ms TTL=253
Reply from 209.165.201.2: bytes=32 time=120ms TTL=253

Ping statistics for 209.165.201.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 120ms, Maximum = 146ms, Average = 133ms

C:\>ssh -l admin 209.165.201.2

Password:
HQ_Router#show users
```

Line	User	Host(s)	Idle	Location
390 vty 0	admin	idle	00:02:05	209.165.201.11
*391 vty 1	admin	idle	00:00:00	

```

    Interface  User          Mode      Idle      Peer Address
HQ_Router#
```

- g. Enter the global configuration mode using **configure terminal** command.
- h. Create an **enable secret** password of **cisco** with the command **enable secret cisco**.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.201.2

Pinging 209.165.201.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 209.165.201.2: bytes=32 time=146ms TTL=253
Reply from 209.165.201.2: bytes=32 time=120ms TTL=253

Ping statistics for 209.165.201.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 120ms, Maximum = 146ms, Average = 133ms

C:\>ssh -l admin 209.165.201.2

Password:
HQ_Router#show users
```

Line	User	Host(s)	Idle	Location
390 vty 0	admin	idle	00:02:05	209.165.201.11
*391 vty 1	admin	idle	00:00:00	

```

    Interface  User          Mode      Idle      Peer Address
HQ_Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HQ_Router(config)#enable secret cisco
HQ_Router(config)#
```

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 1: Send email between users	Step 2	2	
	Step 3	2	
Part 2: Upload and download files using FTP	Step 2	2	
	Step 3d	2	
	Step 3f	2	
Part 3: Remotely access an enterprise router using telnet	Step 1	2	
Part 4: Remotely access an enterprise router using SSH	Step 1	2	
Questions		14	
Packet Tracer Score		86	
Total Score		100	

Cisco Packet Tracer - C:/Users/USER/Downloads/Lab Exercise 1.2 - Communicating in a Cyber World.pka - Joseph Louis S. Opinion - 2026-09-15 08:52:54

File Edit Options View Tools Extensions Window Help

Activity Results Time Elapsed: 00:18:35

Congratulations Joseph Louis S. Opinion! You completed the activity.

Overall Feedback Assessment Items Connectivity Tests

Congratulations! You successfully completed the Packet Tracer -Communicating in a Cyber World activity. However, your final score may change based on your answers to the questions in the Instructions. Consult your instructor.